CSE400 Lecture 4 Scribe

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Assume you are expert scribe generator. Generate a lecture scribe suitable for exam preparation for CSE400 Lecture 4: Joint Probability and Conditional Probability. While writing the scribe, do reason step by step through definitions, assumptions, proofs, derivations, and worked example solutions exactly as presented in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. Use only the provided lecture slides and relevant textbook excerpts. Don't mention any material, explanations or examples that are not present in the context.
Organize the scribe clearly in a formal academic style.

Below is an **exam-preparation lecture scribe** for **CSE400 Lecture 4: Joint Probability and Conditional Probability**, written **strictly from the provided lecture slides**.
I am not adding any extra definitions, theorems, or examples beyond what is present in the context.

CSE400 Lecture 4 Scribe

Joint Probability and Conditional Probability

Course: CSE400 – Fundamentals of Probability in Computing
Lecture Title: Joint Probability and Conditional Probability
Instructor: Dhaval Patel, PhD
Date: January 15, 2026

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1. Introduction to Probability Theory

Lecture 4 builds on the fundamental concepts of probability introduced earlier.

The outline of the lecture includes:

- Experiments, Sample Space, and Events

- Axioms of Probability
- Corollaries and Propositions
- Assigning Probability
- Joint Probability
- Conditional Probability (mentioned in outline)

This lecture focuses mainly on **Joint Probability** and introduces **Conditional Probability** as the next topic.

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2. Experiments, Sample Space, and Events

Probability theory begins with the description of an **experiment**.

2.1 Experiment

Definition:
An **Experiment (E)** is a procedure we perform that produces some result.

Example: Tossing a coin five times is an experiment denoted by E_5.

2.2 Outcome

Definition:
An **Outcome** is a possible result of an experiment.

Example:
One possible outcome of E_5 is:

$$\xi_1 = HHTHT$$

2.3 Event

Definition:
An **Event** is a certain set of outcomes of an experiment.

Example:
For experiment E_5, define event C:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

2.4 Sample Space

Definition:
The **Sample Space (S)** is the collection (set) of **all possible distinct outcomes** of an experiment.

Important properties:

- **Mutually Exclusive:** You can get heads or tails, but not both.
- **Collectively Exhaustive:** You cannot get anything other than heads or tails.

Example: Flipping a coin.

Sample Space Types

The sample space may be:

- Discrete
- Countably infinite
- Continuous

Examples given:

- Flipping a fair coin once
- Rolling a cubical die
- Rolling two dice
- Flipping until tails occurs
- Random number generator on interval $[0, 1)$

---

3. Axioms of Probability

After defining events, the next step is assigning probabilities.

3.1 Probability Definition

Probability is:

- A measure of the likelihood of events
  OR
- A function that assigns a numerical quantity measuring the likelihood of an event

---

3.2 Probability Axioms

The axioms are self-evident statements requiring no proof.

Axiom 1

For any event A:

$$0 \leqslant Pr(A) \leqslant 1$$

Axiom 2

If S is the sample space:

$$Pr(S) = 1$$

Axiom 3 (Additivity for Mutually Exclusive Events)

If:

$$A \cap B = \emptyset$$

Then:

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For an infinite number of mutually exclusive sets:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

---

4. Corollary from Axioms

Corollary 2.1

For a finite number M of mutually exclusive sets:

$$Pr\left(\bigcup A_i\right) = \sum Pr(A_i)$$

The lecture notes state:

- Axiom 3 is equivalent to Corollary 2.1 for finite sample spaces.
- Axiom 3 is necessary for infinite sample spaces.

---

5. Propositions from Probability Axioms

Several propositions follow directly from the axioms.

Proposition 2.1 (Complement Rule)

$$Pr(A^c) = 1 - Pr(A)$$

Proposition 2.2 (Monotonicity)

If:

$$A \subset B$$

Then:

$$Pr(A) \leqslant Pr(B)$$

Proposition 2.3 (Union of Two Sets)

For any sets A and B (not necessarily mutually exclusive):

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Proposition 2.4 (Inclusion–Exclusion for Multiple Events)

For M events:

$$Pr(A_1 \cup A_2 \cup \ldots \cup A_M)$$

equals alternating sums of intersections:

$$\sum Pr(A_i) - \sum Pr(A_i A_j) + \ldots + (-1)^{M+1} Pr(A_1 A_2 \ldots A_M)$$

---

6. Assigning Probabilities

Two approaches are mentioned:

6.1 Classical Approach

Examples:

- Coin flipping: compute $Pr(H)$, $Pr(T)$
- Dice rolling: compute probability of even number

- Pair of dice: probability sum equals five

6.2 Relative Frequency Approach

Example: Pair of dice, sum equals five.

Simulation results show:

$$\frac{n_A}{n} \to Pr(A)$$

Inference:

- Exact probability requires infinite repetitions (drawback)
- Many phenomena are not repeatable

---

7. Joint Probability

This is the main focus of Lecture 4.

7.1 Motivation

Not all events are mutually exclusive.

We are interested in the probability of the intersection:

$$A \cap B$$

7.2 Notation

Joint probability is denoted as:

$$Pr(A, B) \quad \text{or} \quad Pr(A \cap B)$$

7.3 Joint Probability of Multiple Events

For events $A_1, A_2, \ldots, A_M$:

$$Pr(A_1, A_2, \ldots, A_M)$$

7.4 Calculation Approaches

Approach 1: Classical

Step-by-step reasoning:

1. Express events A and B in terms of atomic outcomes.
2. Identify outcomes common to both events.
3. Add the probabilities of those common outcomes.

Approach 2: Relative Frequency

Let n_{AB} be the number of times A and B occur simultaneously in n trials.

$$Pr(A, B) = \lim_{n \to \infty} \frac{n_{AB}}{n}$$

---

8. Worked Examples on Joint Probability

Example 1: Card Deck Example

A standard deck has 52 cards.

Define events:

- A = Red card selected
- B = Number card selected (Ace is a number card)
- C = Heart card selected

Find:

$$Pr(A), Pr(B), Pr(C), Pr(A, B), Pr(A, C), Pr(B, C)$$

Solution Step-by-Step

Individual Probabilities

Step 1: Compute $Pr(A)$

There are 26 red cards (13 hearts + 13 diamonds):

$$Pr(A) = \frac{26}{52}$$

Step 2: Compute $Pr(B)$

There are 40 number cards:

$$Pr(B) = \frac{40}{52}$$

Step 3: Compute $Pr(C)$

There are 13 hearts:

$$Pr(C) = \frac{13}{52}$$

Joint Probabilities

Step 4: Compute $Pr(A, B)$

Red number cards = 10 in hearts + 10 in diamonds:

$$Pr(A, B) = \frac{20}{52}$$

Step 5: Compute $Pr(A, C)$

All hearts are red:

$$Pr(A, C) = \frac{13}{52}$$

Step 6: Compute $Pr(B, C)$

Number cards that are hearts = 10:

$$Pr(B, C) = \frac{10}{52}$$

Example 2: Costume Party Example

Scenario:

Alex has:

- 3 white t-shirts
- 1 radioactive green cape
- 2 pajama pants
- 4 boxer shorts

Question: Probability that Alex selects:

- The green cape
  AND
- Polka-dot boxers

Options: 1/2, 1/4, 1/6

Step 1: Analyze Tops

Total tops:

$$3 + 1 = 4$$

Probability of selecting cape:

$$Pr(\text{Cape}) = \frac{1}{4}$$

(The slide context stops here, so no further steps are included.)

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9. Conditional Probability (Lecture Outline Mention)

The lecture outline lists Conditional Probability with:

- Motivation, notation, and concepts
- Example 3: Cards Without Replacement
- Example 4: Game of Poker
- Example 5: The Missing Key

However, the provided context does not include the actual slides or derivations for these sections.
Therefore, no additional conditional probability definitions or worked examples can be written here without adding outside material.

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10. Exam Revision Key Points

From Lecture 4, students must remember:

- Joint probability is the probability of intersection:

$$Pr(A, B) = Pr(A \cap B)$$

- Events need not be mutually exclusive.
- Two calculation approaches:
  - Classical (atomic outcomes)
  - Relative frequency (limit of repetitions)

- Card deck example is the main worked numerical joint probability problem.

End of Lecture 4 Scribe

If you upload the remaining slides/pages for **Conditional Probability examples (3–5)**, I can generate the full Lecture 4 scribe including those derivations exactly as in the lecture.

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Below is a **formal, exam-oriented lecture scribe** for **CSE400 Lecture 4: Joint Probability and Conditional Probability**, written **strictly and only** from the provided lecture slide context.

All reasoning is presented step-by-step exactly in the logical flow of the slides. No external definitions, formulas, or examples are included beyond the context.

CSE400 Lecture 4 Scribe

Joint Probability and Conditional Probability

Course: CSE400 – Fundamentals of Probability in Computing
Lecture: 4
Topic: Joint Probability and Conditional Probability
Instructor: Dhaval Patel, PhD
Date: January 15, 2026

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1. Lecture Outline

This lecture continues the foundation of probability theory and introduces joint probability.

Topics covered:

1. Introduction to Probability Theory
2. Experiments, Sample Space, and Events
3. Axioms of Probability
4. Corollaries and Propositions
5. Assigning Probabilities
6. Joint Probability
    - Motivation, Notation, Concepts
    - Example 1: Card Deck
    - Example 2: Costume Party
7. Conditional Probability (listed next in outline)
    - Motivation, Notation, Concepts
    - Example 3: Cards Without Replacement
    - Example 4: Game of Poker
    - Example 5: Missing Key

---

2. Introduction to Probability Theory

Probability theory begins with understanding random experiments and events.

---

3. Experiments, Sample Space, and Events

3.1 Experiment

Definition:

An **Experiment (E)** is a procedure we perform that produces some result.

Example:

Tossing a coin five times is an experiment denoted by:

$$E_5$$

3.2 Outcome

Definition:

An **Outcome** is a possible result of an experiment.

Example:

One outcome of E_5 is:

$$\xi_1 = HHTHT$$

3.3 Event

Definition:

An **Event** is a certain set of outcomes of an experiment.

Example:

Event C for experiment E_5:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

3.4 Sample Space

Definition:
The **Sample Space (S)** is the collection of all possible distinct outcomes of an experiment.

The outcomes must be:

- **Mutually Exclusive:** cannot occur together
- **Collectively Exhaustive:** cover all possibilities

Example: Flip a coin

- Heads or tails, not both
- Nothing else is possible

Sample Space Types

Sample space may be:

- Discrete
- Countably infinite
- Continuous

Examples:

- Flipping a fair coin once
- Rolling a cubical die
- Rolling two dice
- Flipping until tails occurs
- Random number generator in $[0, 1)$

---

4. Axioms of Probability

After defining events, we assign probabilities.

4.1 Probability Meaning

Probability is:

- A measure of likelihood of events

  OR
- A function producing a numerical quantity measuring likelihood

---

4.2 Probability Axioms

Axioms are self-evident and require no proof.

Axiom 1: Bounds

For any event A:

$$0 \leqslant Pr(A) \leqslant 1$$

Axiom 2: Certain Event

For sample space S:

$$Pr(S) = 1$$

Axiom 3: Additivity for Mutually Exclusive Events

If:

$$A \cap B = \varnothing$$

Then:

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For infinitely many mutually exclusive sets A_i:

$$Pr(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} Pr(A_i)$$

---

5. Corollary from Probability Axioms

Corollary 2.1

For a finite number M of mutually exclusive sets:

$$Pr(\bigcup A_i) = \sum Pr(A_i)$$

Notes:

- Equivalent to Axiom 3 when sample space is finite
- Generality of Axiom 3 is necessary for infinite sample spaces

---

6. Propositions from Probability Axioms

Proposition 2.1: Complement Rule

$$Pr(A^c) = 1 - Pr(A)$$

Proposition 2.2: Subset Rule

If:

$$A \subset B$$

Then:

$$Pr(A) \leqslant Pr(B)$$

Proposition 2.3: Union Rule

For any sets A and B:

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Proposition 2.4: Inclusion–Exclusion

For events $A_1, \ldots, A_M$:

$$Pr(A_1 \cup A_2 \cup \ldots \cup A_M)$$

is computed by alternating sums of intersections:

$$\sum Pr(A_i) - \sum Pr(A_i A_j) + \ldots + (-1)^{M+1} Pr(A_1 A_2 \ldots A_M)$$

---

7. Assigning Probabilities

7.1 Classical Approach

Examples listed:

- Coin flipping: compute $Pr(H)$, $Pr(T)$
- Dice rolling: probability of even number
- Pair of dice: probability sum equals five

7.2 Relative Frequency Approach

Example: Pair of dice, sum equals five.

Simulation table shows:

$$\frac{n_A}{n}$$

approaches probability as trials increase.

Inference:

- Exact probability requires infinite repetitions
- Many random phenomena are not repeatable

---

8. Joint Probability

This is the central topic of Lecture 4.

8.1 Motivation

All events are not mutually exclusive.

We want the probability of intersection:

$$A \cap B$$

8.2 Notation

Joint probability is denoted as:

$$Pr(A, B) \quad \text{or} \quad Pr(A \cap B)$$

8.3 Joint Probability of Multiple Events

For events $A_1, A_2, \dots, A_M$:

$$Pr(A_1, A_2, \dots, A_M)$$

8.4 Calculation Approaches

Approach 1: Classical

Step-by-step:

1. Express events A and B in terms of atomic outcomes
2. Write the set of outcomes common to both
3. Calculate probability of each common outcome

Approach 2: Relative Frequency

Let n_{AB} be the number of times A and B occur simultaneously in n trials:

$$Pr(A, B) = \lim_{n \to \infty} \frac{n_{AB}}{n}$$

9. Worked Examples

Example 1: Card Deck Example

A standard deck has 52 cards.

Define:

- A = {Red card selected}
- B = {Number card selected (Ace is a number card)}
- C = {Heart card selected}

Find:

$$Pr(A), Pr(B), Pr(C), Pr(A, B), Pr(A, C), Pr(B, C)$$

Solution

Step 1: Individual Probabilities

Probability of Red Card
There are 26 red cards:

- 13 hearts
- 13 diamonds

$$Pr(A) = \frac{26}{52}$$

Probability of Number Card
There are 40 number cards:

$$Pr(B) = \frac{40}{52}$$

Probability of Heart Card
There are 13 hearts:

$$Pr(C) = \frac{13}{52}$$

Step 2: Joint Probabilities

Joint Probability $Pr(A, B)$
There are:

- 10 red number cards in hearts

- 10 red number cards in diamonds

Total = 20

$$Pr(A, B) = \frac{20}{52}$$

Joint Probability $Pr(A, C)$
All hearts are red:

$$Pr(A, C) = \frac{13}{52}$$

Joint Probability $Pr(B, C)$
Number cards that are hearts = 10:

$$Pr(B, C) = \frac{10}{52}$$

Example 2: Costume Party Example

Alex has:

- 3 food-stained white t-shirts
- 1 radioactive green cape
- 2 dinosaur pajama pants
- 4 polka-dot boxer shorts

Question:

Probability outfit is:

- Green cape
  AND
- Polka-dot boxers

Options:

A) 1/2
B) 1/4
C) 1/6

Solution (as provided)

Step 1: Analyze Tops

Total tops:

$$3 + 1 = 4$$

Probability of cape:

$$Pr(\text{Cape}) = \frac{1}{4}$$

(The provided slide context stops here, so no further steps can be included.)

---

10. Conditional Probability

The lecture outline lists Conditional Probability next, with examples:

- Cards Without Replacement
- Game of Poker
- Missing Key

However, the provided slide context does not include the actual definitions or worked solutions for these sections.

Therefore, no conditional probability derivations can be written without adding external material.

---

11. Exam Revision Summary

Key points from Lecture 4:

- Joint probability measures intersection:

$$Pr(A, B) = Pr(A \cap B)$$

- Events need not be mutually exclusive
- Two approaches:
  - Classical (atomic outcomes)
  - Relative frequency (limit form)
- Card deck example is the main complete numerical example

End of Lecture 4 Scribe

If you upload the remaining pages/slides containing the full **Conditional Probability** section (Examples 3–5), I can generate the complete lecture scribe including those exact derivations.

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CSE400 Lecture 4 Scribe

Joint Probability and Conditional Probability

Course: CSE400 – Fundamentals of Probability in Computing
Lecture Number: 4
Lecture Title: Joint Probability and Conditional Probability
Instructor: Dhaval Patel, PhD
Date: January 15, 2026

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1. Lecture Outline

The lecture develops probability theory from basic foundations and then introduces joint probability.

Topics covered:

- Introduction to Probability Theory
- Experiments, Sample Space, and Events
- Axioms of Probability
- Corollaries and Propositions from Probability Axioms
- How to Assign Probability
  - Classical Approach
  - Relative Frequency Approach
- Joint Probability
  - Motivation, Notation, and Concepts
  - Example 1: Card Deck Example

    - Example 2: Costume Party Example
- Conditional Probability
    - Motivation, Notation, and Concepts
    - Example 3: Cards Without Replacement
    - Example 4: Game of Poker
    - Example 5: The Missing Key

---

2. Introduction to Probability Theory

Probability theory begins with defining experiments, outcomes, events, and sample spaces before assigning probabilities.

---

3. Experiments, Sample Space, and Events

3.1 Experiment

Definition:
An **Experiment (E)** is a procedure we perform that produces some result.

Example:
Tossing a coin five times is denoted as:

$$E_5$$

3.2 Outcome

Definition:
An **Outcome (ξ)** is a possible result of an experiment.

Example:
One outcome of E_5 is:

$$\xi_1 = HHTHT$$

3.3 Event

Definition:
An **Event** is a certain set of outcomes of an experiment.

Example:
Event C for experiment E_5:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

3.4 Sample Space

Definition:
The **Sample Space (S)** is the collection or set of *all possible distinct* outcomes of an experiment.

Key properties:

- **Mutually Exclusive:** outcomes cannot occur together
- **Collectively Exhaustive:** outcomes cover all possibilities

Example: Flip a coin

- Mutually Exclusive: heads or tails, not both
- Collectively Exhaustive: nothing else is possible

Sample Space Types

Sample space may be:

- Discrete
- Countably infinite
- Continuous

Examples listed:

- Flipping a fair coin once
- Rolling a cubical die
- Rolling two dice
- Flip until a tail occurs
- Random number generator in $[0, 1)$

---

4. Axioms of Probability

After defining events, the next step is assigning probabilities.

4.1 Probability Definition

Probability is:

- A measure of the likelihood of various events
  OR
- A function of an event that produces a numerical quantity measuring likelihood

---

4.2 Axioms

Axioms are self-evident and require no proof.

Axiom 1 (Bounds)

For any event A:

$$0 \leqslant Pr(A) \leqslant 1$$

Axiom 2 (Certain Event)

If S is the sample space:

$$Pr(S) = 1$$

Axiom 3 (Additivity)

If $A \cap B = \emptyset$:

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For infinitely many mutually exclusive sets A_i:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

---

5. Corollary from Probability Axioms

Corollary 2.1

For a finite number M of mutually exclusive sets:

$$Pr\left(\bigcup A_i\right) = \sum Pr(A_i)$$

Important slide notes:

- Axiom 3 is equivalent to Corollary 2.1 when the sample space is finite
- The generality of Axiom 3 is necessary for infinite sample spaces

---

6. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

$$Pr(A^c) = 1 - Pr(A)$$

Proposition 2.2 (Subset Rule)

If:

$$A \subset B$$

Then:

$$Pr(A) \leqslant Pr(B)$$

Proposition 2.3 (Union Rule)

For any sets A and B:

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Proposition 2.4 (Inclusion–Exclusion Principle)

For events $A_1, \ldots, A_M$:

$$Pr(A_1 \cup A_2 \cup \ldots \cup A_M)$$

is computed by alternating sums of intersections:

$$\sum Pr(A_i) - \sum Pr(A_i A_j) + \ldots + (-1)^{M+1} Pr(A_1 A_2 \ldots A_M)$$

7. Assigning Probabilities

7.1 Classical Approach

Examples listed:

- Coin flipping: compute $Pr(H)$, $Pr(T)$
- Dice rolling: probability of even number
- Pair of dice: probability sum equals five

7.2 Relative Frequency Approach

Example: Pair of dice, sum equals five.

Simulation table shows:

$$\frac{n_A}{n}$$

approaches probability as trials increase.

Inference from slides:

- Exact probability requires infinite repetitions
- Many random phenomena are not repeatable

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8. Joint Probability

8.1 Motivation

All events are not mutually exclusive.

We are interested in the probability of intersection:

$$A \cap B$$

8.2 Notation

Joint probability is denoted as:

$$Pr(A, B) \quad \text{or} \quad Pr(A \cap B)$$

8.3 Multiple Events

For events $A_1, A_2, \dots, A_M$:

$$Pr(A_1, A_2, \dots, A_M)$$

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Approach 1: Classical

Step-by-step reasoning:

1. Express both events A and B using atomic outcomes
2. Write the set of atomic outcomes common to both
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Approach 2: Relative Frequency

Let n_{AB} be the number of times A and B occur simultaneously in n trials:

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Step 1: Individual Probabilities

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- 13 diamonds

$$Pr(A) = \frac{26}{52}$$

Probability of Number Card
There are 40 number cards:

$$Pr(B) = \frac{40}{52}$$

Probability of Heart Card

There are 13 hearts:

$$Pr(C) = \frac{13}{52}$$

Step 2: Joint Probabilities

Joint Probability $Pr(A, B)$
10 number cards per suit.

Red suits: hearts + diamonds:

$$10 + 10 = 20$$

$$Pr(A, B) = \frac{20}{52}$$

Joint Probability $Pr(A, C)$
All hearts are red:

$$Pr(A, C) = \frac{13}{52}$$

Joint Probability $Pr(B, C)$
Number cards that are hearts = 10:

$$Pr(B, C) = \frac{10}{52}$$

Example 2: Costume Party Example

Alex has:

- 3 food-stained white t-shirts
- 1 radioactive green cape
- 2 dinosaur pajama pants
- 4 polka-dot boxer shorts

Probability outfit is:

- Green cape
  AND
- Polka-dot boxers

Options:

A) 1/2
B) 1/4
C) 1/6

Step 1: Analyze Tops

Total tops:

$$3 + 1 = 4$$

Probability of cape:

$$Pr(\text{Cape}) = \frac{1}{4}$$

(The slide context stops at Step 1, so no further solution steps can be added.)

---

10. Conditional Probability

The lecture outline includes Conditional Probability with Examples 3–5, but the provided slide context does not include the actual definitions or worked derivations.

Therefore, conditional probability content cannot be written without introducing material not present in the context.

11. Exam Revision Summary

Key formulas from Lecture 4:

- Joint probability:

$$Pr(A, B) = Pr(A \cap B)$$

- Union probability:

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

- Complement rule:

$$Pr(A^c) = 1 - Pr(A)$$

- Relative frequency definition:

$$Pr(A, B) = \lim_{n \to \infty} \frac{n_{AB}}{n}$$

End of Lecture 4 Scribe

If you upload the remaining slides/pages (covering Conditional Probability Examples 3–5), I can generate the complete lecture scribe for the full lecture exactly as required.